

Amazon ECR (Elastic Container Registry) - Service Overview

Introduction

Amazon Elastic Container Registry (ECR) is a fully managed Docker container registry that makes it easy to store, manage, share, and deploy container images and artifacts. ECR is integrated with Amazon ECS, Amazon EKS, and AWS Lambda, simplifying the development-to-production workflow.

Repositories

An ECR repository is a collection of container images sharing a common name, similar to a Docker Hub repository. Repositories can be private (accessible only to authenticated, authorized principals) or public (accessible to anyone via the Amazon ECR Public Gallery).

Image Tags and Immutability

Images pushed to a repository are identified by a tag (e.g. "myapp:v1.2.0" or "myapp:latest"). ECR supports tag immutability, which prevents image tags from being overwritten once pushed. This is a recommended production setting, since it guarantees that a given tag always refers to exactly one, unchanging image, which is important for reproducible deployments and auditability.

Authentication

Docker clients authenticate to ECR using short-lived authentication tokens obtained via the AWS CLI (`aws ecr get-login-password`) or the AWS SDK. These tokens are valid for 12 hours. In CI/CD pipelines, this authentication step typically occurs immediately before the `docker push` step, using credentials scoped to an IAM role with only `ecr:GetAuthorizationToken`, `ecr:BatchCheckLayerAvailability`, `ecr:PutImage`, and related push permissions.

Image Scanning

ECR provides two types of vulnerability scanning:

- Basic scanning: uses the open-source Clair scanning engine, triggered on image push.
- Enhanced scanning: integrates with Amazon Inspector for continuous, automated vulnerability scanning of both operating system and programming language package vulnerabilities.

Lifecycle Policies

Repositories can have lifecycle policies that automatically expire old or untagged images based on rules such as image count or image age. This helps control storage costs and keeps repositories from accumulating obsolete images indefinitely.

Repository Permissions

Access to ECR repositories is controlled via IAM policies and, optionally, repository policies (resource-based policies similar to S3 bucket policies), which can grant cross-account pull or push access to specific principals.

Integration with ECS and Fargate

When an ECS task definition references a container image, it typically points to an ECR repository URI in the form `<account-id>.dkr.ecr.<region>.amazonaws.com/<repository>:<tag>`. The ECS task execution role must have permission to authenticate to ECR and pull the specified image.

Typical CI/CD Push Sequence

A typical GitHub Actions or similar CI/CD pipeline pushing to ECR follows this sequence: (1) checkout code, (2) configure AWS credentials via a short-lived OIDC-federated role rather than long-lived access key secrets, (3) run `aws ecr get-login-password | docker login`, (4) `docker build` the image with a tag such as the Git commit SHA, (5) `docker push` the tagged image to the repository, and (6) optionally trigger a deployment step (updating an ECS service or Terraform apply) gated behind a manual approval. Tagging by commit SHA, rather than only `latest`, ensures every pushed image is traceable back to the exact source code that produced it.

Replication

ECR supports cross-region and cross-account replication configurations, automatically copying images pushed to a source repository into one or more destination regions or accounts, which is useful for reducing pull latency for services running in multiple regions, or for maintaining a disaster-recovery copy of critical images.

Common Use Cases

- Central image registry for CI/CD pipelines building and deploying containerized applications
- Storing base images used across multiple internal teams or services
- Distributing container images across AWS accounts within an organization
- Feeding container images into ECS, EKS, or Lambda container image deployments

Best Practices

- Enable tag immutability on production repositories.
- Enable image scanning to catch known vulnerabilities before deployment.
- Apply lifecycle policies to prevent unbounded storage growth from CI pipeline builds.

- Scope the IAM role used by CI/CD to push images only to the specific repositories it needs, not account-wide ECR access.